

ENTERPRISE MULTI-PROTOCOL ROUTING CHALLENGE

Design, Implementation, and Failover Engineering in Mixed OSPF/RIPv2 Environments



NETWORK ADMINISTRATION

ITSimplera Institute

INTERNSHIP PROJECT REPORT

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Title:	Enterprise Multi-Protocol Routing Challenge Week 3
Submission Date:	July 16, 2026

Technical Glossary of Terms

To ensure clarity throughout this design dossier, the fundamental networking protocols and architectural concepts used are defined below:

1. Open Shortest Path First (OSPF)

A link-state routing protocol designed for IP networks, operating inside a single Autonomous System (AS). OSPF uses Dijkstra's Shortest Path First (SPF) algorithm to calculate the most efficient path to a destination based on cumulative interface bandwidth (cost). It is highly scalable, supports rapid convergence, and is typically deployed in medium-to-large enterprise cores.

2. Routing Information Protocol version 2 (RIPv2)

A distance-vector routing protocol that uses hop-count as its primary metric to determine the best path to a network segment. RIPv2 is classless (supports Variable Length Subnet Masking - VLSM) and sends routing updates via multicast (224.0.0.9). It has a maximum hop-count of 15, making any destination 16 hops away unreachable, which limits its deployment to smaller branch architectures.

3. Autonomous System Boundary Router (ASBR)

A router that sits at the edge of one or more distinct routing domains (such as OSPF and RIPv2) and is responsible for introducing routing information learned from external networks into the local routing domain. In this project, `BoundaryR3` acts as the ASBR.

4. Route Redistribution

The process of taking routing information from one routing protocol (e.g., OSPF) and importing/translating it into another protocol (e.g., RIPv2). Because different protocols use completely incompatible metrics (bandwidth cost vs. hops), the ASBR must translate these values into a metric the receiving protocol can understand.

5. Floating Static Route

A manually configured backup static route that is assigned an elevated Administrative Distance (AD). Because routers prioritize routes with lower AD values, this backup route remains completely inactive (dormant) in the routing table unless the primary, dynamically learned routing path fails.

6. Administrative Distance (AD)

A rating of the trustworthiness of a routing information source, expressed as an integer value from 0 to 255. A lower value indicates a more trusted protocol. For reference in this project:

- **Directly Connected Interface:** 0
- **Static Route:** 1
- **OSPF:** 110
- **RIPv2:** 120
- **Our Floating Static Route:** 50 (specifically configured to override dynamic protocol metrics during failover)

Section 1 — Executive Summary & Project Scope

1.1 Executive Summary

In the modern enterprise landscape, corporate expansions, mergers, and acquisitions regularly force IT departments to integrate separate network infrastructures. This project addresses a real-world corporate merger scenario: connecting the **Corporate Headquarters** which utilizes a modern, highly scalable **Open Shortest Path First (OSPF)** link-state routing protocol with a newly **Acquired Branch Office** running a legacy **Routing Information Protocol version 2 (RIPv2)** system.

The primary objective of this engineering task is to establish complete, seamless bidirectional IP communication across these distinct routing domains. By utilizing **mutual route redistribution** at the autonomous system boundary router (ASBR), we bridge the gap between OSPF's cost-based metrics and RIPv2's hop-count metrics. Furthermore, to guarantee uninterrupted business continuity, a **Floating Static Route** with an elevated Administrative Distance of 50 has been configured over a secondary backup WAN link. This ensures automatic path failover in the event of a primary high-speed fiber link failure.

1.2 Project Scope & Boundaries

This deployment is bounded by the following design constraints and requirements:

- **Multi-Area OSPF Segmentation:** The Core network must be divided into OSPF Area 0 (Backbone) and OSPF Area 1 (Transit) to restrict link-state advertisement (LSA) flooding and optimize router CPU cycles.
- **RIPv2 Modernization:** The remote branch must run RIPv2 with classless routing updates enabled to support variable-length subnet masking (VLSM).

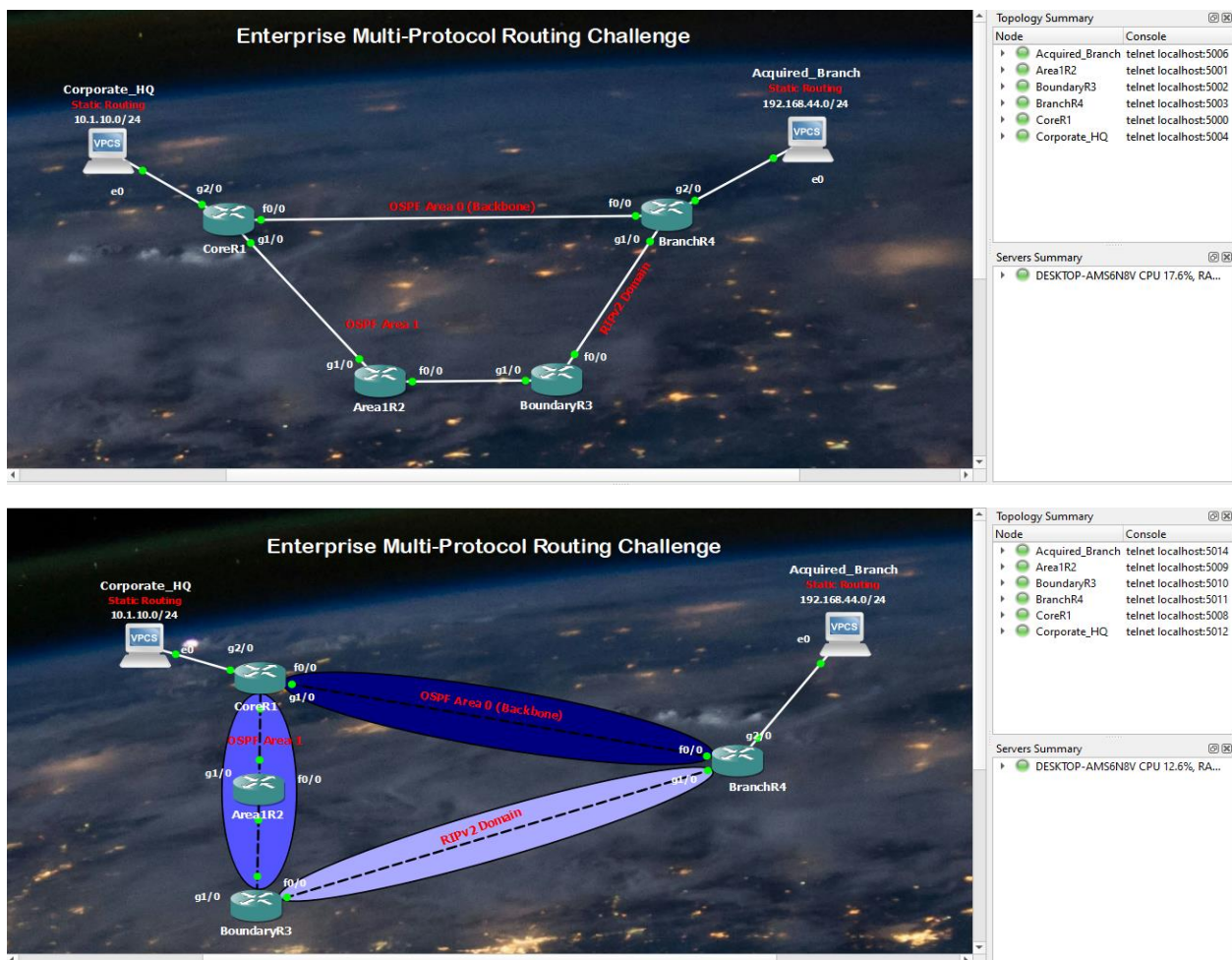
- **Controlled Route Leakage:** Mutual redistribution must be carefully configured on the ASBR (BoundaryR3) to prevent routing loops or suboptimal routing paths.
- **Deterministic Failover:** The backup link must remain strictly idle during normal operations and dynamically assume routing responsibility only when the primary path is physically or logically down.

SECTION 2 — NETWORK TOPOLOGY & SEGMENT ARCHITECTURE

2.1 Logical Topology Design

The network topology designed for this Enterprise Routing Challenge represents a multi-tiered, multi-protocol enterprise network architecture. The layout simulates a corporate headquarters transitioning into an acquired branch network via an intermediate boundary layer.

The topology is logically segmented into three distinct routing domains to isolate local routing updates, optimize CPU utilization on network nodes, and establish clear administrative control boundaries:



2.2 Network Nodes & Their Architectural Roles

To bridge the gap between distinct network segments, each router in the topology is assigned a specific structural role that defines how it processes, filters, and propagates routing information. The design relies on a multi-tiered hierarchical architecture, separating backbone transit traffic from edge distribution and translation boundaries.

1. CoreR1 (OSPF Area 0 Backbone Router)

- **Role:** Serves as the primary core router for the Corporate Headquarters.
- **Interfaces:**
 - GigabitEthernet1/0 (IP: 10.1.1.1/24) connecting to the Backbone Area (Area 0).
 - GigabitEthernet2/0 (IP: 10.1.10.1/24) acting as the gateway for the Corporate Headquarters local LAN segment.
 - FastEthernet0/0 (IP: 192.168.14.1/24) configured as the exit interface for the Floating Static Backup Path connecting directly to BranchR4.

2. Area1R2 (OSPF Transit Area 1 Router)

- **Role:** Acts as an internal Area 1 router helping to segment OSPF traffic and minimize SPF calculations.
- **Interfaces:**
 - GigabitEthernet1/0 (IP: 10.1.1.2/24) pointing north to the OSPF Backbone Area (Area 0).
 - FastEthernet0/0 (IP: 10.1.2.1/24) pointing south to the OSPF Area 1 transit network segment connecting to BoundaryR3.

3. BoundaryR3 (Autonomous System Boundary Router - ASBR)

- **Role:** Acts as the translation gateway (ASBR) between OSPF and RIPv2. This device handles mutual route redistribution, importing OSPF metrics into RIPv2 and vice-versa.
- **Interfaces:**
 - FastEthernet0/0 (IP: 10.1.2.3/24) running OSPF Area 1 to talk to the corporate network.
 - GigabitEthernet1/0 (IP: 192.168.34.3/24) running RIPv2 to talk to the Acquired Branch.

4. BranchR4 (RIPv2 Remote Branch Router)

- **Role:** Represents the gateway of the newly acquired remote branch office, operating legacy RIPv2.

- **Interfaces:**
 - GigabitEthernet1/0 (IP: 192.168.34.4/24) running RIPv2 to receive redistributed enterprise routes from BoundaryR3.
 - GigabitEthernet2/0 (IP: 192.168.44.1/24) acting as the gateway for the remote Branch LAN segment.
 - FastEthernet0/0 (IP: 192.168.14.4/24) acting as the termination point for the backup floating static link coming from CoreR1.

2.3 Physical and Logical Segments

- **Corporate Core Segment (10.1.1.0/24):** High-speed Backbone Area 0 link linking CoreR1 and Area1R2.
- **Transit Area 1 Segment (10.1.2.0/24):** Mid-tier Area 1 distribution network connecting Area1R2 and BoundaryR3.
- **Translation Link (192.168.34.0/24):** The boundaries of translation between OSPF and RIPv2 controlled by BoundaryR3.
- **Resilient Backup Segment (192.168.14.0/24):** A secondary path configured as a slow serial/FastEthernet backup link to provide fallback connectivity during a primary link failure.

3.2 IP Addressing Table

The table below lists every interface, its logical purpose, assigned IP address, and subnet mask across the entire multi-protocol topology:

Device Name	Interface ID	IP Address	Subnet Mask	Logical Network Segment / Purpose
CoreR1	Gi1/0	10.1.1.1	255.255.255.0	OSPF Area 0 Backbone Link (HQ Core)
CoreR1	Gi2/0	10.1.10.1	255.255.255.0	Corporate HQ LAN Gateway

Device Name	Interface ID	IP Address	Subnet Mask	Logical Network Segment / Purpose
CoreR1	Fa0/0	192.168.14.1	255.255.255.0	Resilient Backup Path to Branch
Area1R2	Gi1/0	10.1.1.2	255.255.255.0	OSPF Area 0 / Area 1 Transition Link
Area1R2	Fa0/0	10.1.2.1	255.255.255.0	OSPF Area 1 Transit Network Segment
BoundaryR3	Fa0/0	10.1.2.3	255.255.255.0	OSPF Area 1 Boundary Interface
BoundaryR3	Gi1/0	192.168.34.3	255.255.255.0	RIPv2 Redistribution Boundary Interface
BranchR4	Gi1/0	192.168.34.4	255.255.255.0	RIPv2 Segment Link
BranchR4	Gi2/0	192.168.44.1	255.255.255.0	Acquired Branch LAN Gateway
BranchR4	Fa0/0	192.168.14.4	255.255.255.0	Backup Link Termination Interface

3.3 Subnet Isolation Analysis

To maintain traffic control and keep routing updates predictable, the subnets are partitioned into distinct zones:

1. The Core Enterprise Zone (10.1.0.0/16 range)

- **10.1.1.0/24:** Dedicated strictly to OSPF Area 0. This is the backbone of the enterprise corporate office where core control plane traffic travels.
- **10.1.10.0/24:** Reserved for the local HQ client LAN, keeping user data traffic isolated from router-to-router control links.
- **10.1.2.0/24:** Acts as the OSPF Area 1 transit path, ensuring that any routing updates originating from the boundary router must step through Area 1 before reaching the backbone.

2. The Acquired Branch Zone (192.168.0.0/16 range)

- **192.168.34.0/24:** Dedicated to the primary routing interface between the translation gateway and the branch.
- **192.168.44.0/24:** Houses the remote branch LAN users, separating branch operations from the transit link.
- **192.168.14.0/24:** Reserved exclusively for the low-speed backup link. This link remains quiet and carries zero user traffic under normal, healthy network conditions.

4. Core OSPF Multi-Area Implementation

4.1 Multi-Area Design & Logic

To reduce overall routing database size, the corporate headquarters is structured into OSPF Area 0 and OSPF Area 1. This hierarchical segmentation restricts routing updates, limiting Dijkstra calculations to their respective areas during a network topology change.

To mitigate this scaling issue, a hierarchical, two-tier **Multi-Area OSPF architecture** was engineered for this project:

- **Area 0 (The Backbone):** Acting as the central transit core of the enterprise, Area 0 (comprising CoreR1 and the backbone-facing interface of Area1R2) is responsible for fast, reliable packet forwarding. By design, all inter-area traffic must traverse the backbone.
- **Area 1 (The Transit Domain):** Acting as a non-backbone area, Area 1 (comprising BoundaryR3 and the downstream-facing interface of Area1R2) localizes routing complexities.

4.2 Core Router 1 (CoreR1) OSPF Configuration

CoreR1 serves as the central hub of the enterprise backbone network. It maintains high-speed link-state adjacencies exclusively within OSPF Area 0, routing backbone traffic and propagating inter-area prefixes.

- **CoreR1# configure terminal**
- interface GigabitEthernet1/0
- ip address 10.1.1.1 255.255.255.0
- negotiation auto
- no shutdown
- interface Loopback0
- description Core Router ID Loopback
- ip address 1.1.1.1 255.255.255.255
- router ospf 1
- router-id 1.1.1.1
- network 10.1.1.0 0.0.0.255 area 0
- network 1.1.1.1 0.0.0.0 area 0
- exit
- end
- **CoreR1# write memory**

4.3 Area1R2 Configuration (Area Border Router - ABR)

Area1R2 operates as the critical Area Border Router (ABR). It splits its link-state database: GigabitEthernet1/0 connects to CoreR1 in Area 0, while FastEthernet0/0 connects to BoundaryR3 in Area 1, isolating the backbone from local Area 1 flooding.

- **Area1R2# configure terminal**
- interface GigabitEthernet1/0
- ip address 10.1.1.2 255.255.255.0
- negotiation auto
- no shutdown
- interface FastEthernet0/0
- ip address 10.1.2.1 255.255.255.0
- no shutdown
- router ospf 1
- router-id 2.2.2.2
- network 10.1.1.0 0.0.0.255 area 0
- network 10.1.2.0 0.0.0.255 area 1
- exit
- end
- **Area1R2# write memory**

4.4 BoundaryR3 Configuration (Autonomous System Boundary Router - ASBR)

BoundaryR3 functions as the Autonomous System Boundary Router (ASBR), establishing the gateway between the OSPF core and the RIPv2 legacy branch domain. It features full mutual redistribution between OSPF 1 and RIP processes.

- **BoundaryR3# configure terminal**
- interface GigabitEthernet1/0
- ip address 10.1.2.3 255.255.255.0
- negotiation auto
- no shutdown
- interface FastEthernet0/0
- ip address 192.168.34.3 255.255.255.0
- no shutdown
- router ospf 1
- router-id 192.168.34.3
- network 10.1.2.0 0.0.0.255 area 1
- redistribute rip subnets
- exit
- router rip
- version 2
- network 192.168.34.0
- redistribute ospf 1 metric 5
- no auto-summary
- exit
- end
- **BoundaryR3# write memory**

4.5 BranchR4 Configuration (Legacy Branch Edge Router)

BranchR4 operates as the gateway for the remote branch subnets. It runs lightweight RIPv2 classless dynamic routing on its local interfaces, receiving the corporate enterprise routing table directly via BoundaryR3.

- **BranchR4# configure terminal**
- interface GigabitEthernet1/0
- ip address 192.168.34.4 255.255.255.0
- negotiation auto
- no shutdown
- interface GigabitEthernet2/0
- ip address 192.168.44.1 255.255.255.0
- negotiation auto
- no shutdown
- router rip
- version 2
- network 192.168.34.0

- network 192.168.44.0
- no auto-summary
- exit
- end
- **BranchR4# write memory**

5. Diagnostic & Verification Commands Reference

To validate the operational state of the multi-protocol routing architecture and ensure loop-free end-to-end reachability, a suite of standardized Cisco IOS verification commands is utilized. These diagnostic tools allow for direct inspection of protocol behaviour, neighbour states, and routing tables across the OSPFv2 and RIPv2 domains without altering the active configurations.

5.1 OSPF Core and Backbone Configuration

```
CoreR1
*Jul 13 23:39:03.343: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jul 13 23:39:03.351: %LINK-3-UPDOWN: Interface GigabitEthernet1/0, changed state to up
*Jul 13 23:39:03.355: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Jul 13 23:39:04.375: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Jul 13 23:39:04.379: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up
*Jul 13 23:39:04.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
*Jul 13 23:39:08.731: %SYS-5-CONFIG_I: Configured from memory by console
*Jul 13 23:39:09.879: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.3(3)XB12, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Tue 19-Nov-13 04:39 by prod_rel_team
*Jul 13 23:39:09.987: %SNMP-5-COLDSTART: SNMP agent on host CoreR1 is undergoing a cold start
*Jul 13 23:39:10.747: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jul 13 23:39:10.751: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Jul 13 23:39:20.559: %SYS-3-CPUHOG: Task is running for (2004)msecs, more than (2000)msecs (3/3), process = Crypto CA.
-Traceback= 0x64CB7F80z
*Jul 13 23:39:50.351: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet1/0 from LOADING to FULL, Loading Done
CoreR1#
CoreR1#
CoreR1#show ip ospf neighbor
      ^
% Invalid input detected at '^' marker.

CoreR1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
10.2.2.2       1     FULL/BDR        00:00:14    10.1.1.2       GigabitEthernet1/0
CoreR1#
```

```
Area1R2
Area1R2(config-router)#
*Jul 16 15:15:16.963: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet1/0 from LOADING to FULL, Loading Done
Area1R2(config-router)#end
Area1R2#wr
*Jul 16 15:15:27.895: %SYS-5-CONFIG_I: Configured from console by console
Area1R2#write memory
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
Area1R2#
Area1R2#
Area1R2#
Area1R2#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.2.2.2
  It is an area border router
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.1.1.0 0.0.0.255 area 0
    10.2.2.0 0.0.0.255 area 1
  --More--
```

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5.2 Autonomous System Boundary Router (ASBR) & Redistribution

```
BoundaryR3
*Jul 13 23:39:09.811: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.3(3)XB12
, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Tue 19-Nov-13 04:39 by prod_rel_team
*Jul 13 23:39:09.899: %SNMP-5-COLDSTART: SNMP agent on host BoundaryR3 is underg
oing a cold start
*Jul 13 23:39:10.475: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jul 13 23:39:10.479: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Jul 13 23:39:54.835: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet
1/0 from LOADING to FULL, Loading Done
BoundaryR3#
BoundaryR3#
BoundaryR3#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.34.3
  It is an autonomous system boundary router
  Redistributing External Routes from,
    rip, includes subnets in redistribution
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  --More--
```

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```
BranchR4
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Tue 19-Nov-13 04:39 by prod_rel_team
*Jul 13 23:39:10.031: %SNMP-5-COLDSTART: SNMP agent on host BranchR4 is undergoing a cold start
*Jul 13 23:39:10.587: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jul 13 23:39:10.587: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
BranchR4#
BranchR4#show ip ospf neighbor
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 3 subnets
R    10.1.1.0 [120/5] via 192.168.34.3, 00:00:06, GigabitEthernet1/0
R    10.1.10.0 [120/5] via 192.168.34.3, 00:00:06, GigabitEthernet1/0
R    10.2.2.0 [120/5] via 192.168.34.3, 00:00:06, GigabitEthernet1/0
    192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/0
L    192.168.14.4/32 is directly connected, FastEthernet0/0
    192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet1/0
L    192.168.34.4/32 is directly connected, GigabitEthernet1/0
--More--
```

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```
CoreR1
CoreR1#show ip ospf neighbor
% Invalid input detected at '^' marker.

CoreR1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.2.2.2          1    FULL/BDR        00:00:14    10.1.1.2       GigabitEtherne
t1/0
CoreR1#
CoreR1#
CoreR1#
CoreR1#
CoreR1#
CoreR1#
CoreR1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet1/0
L    10.1.1.1/32 is directly connected, GigabitEthernet1/0
C    10.1.10.0/24 is directly connected, GigabitEthernet2/0
L    10.1.10.1/32 is directly connected, GigabitEthernet2/0
O IA  10.2.2.0/24 [110/2] via 10.1.1.2, 00:26:48, GigabitEthernet1/0
    192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/0
L    192.168.14.1/32 is directly connected, FastEthernet0/0
O E2  192.168.34.0/24 [110/20] via 10.1.1.2, 00:26:48, GigabitEthernet1/0
--More--
```

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```
Area1R2
*Jul 13 23:39:08.135: %SYS-5-CONFIG_I: Configured from memory by console
*Jul 13 23:39:09.399: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.3(3)XB12
, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Tue 19-Nov-13 04:39 by prod_rel_team
*Jul 13 23:39:09.483: %SNMP-5-COLDSTART: SNMP agent on host Area1R2 is undergoin
g a cold start
*Jul 13 23:39:09.983: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jul 13 23:39:09.983: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Jul 13 23:39:50.323: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.14.1 on GigabitEthe
rnet1/0 from LOADING to FULL, Loading Done
*Jul 13 23:39:54.651: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.34.3 on FastEthern
et0/0 from LOADING to FULL, Loading Done
Area1R2#
Area1R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet1/0
L    10.1.1.2/32 is directly connected, GigabitEthernet1/0
O    10.1.10.0/24 [110/2] via 10.1.1.1, 00:19:12, GigabitEthernet1/0
C    10.2.2.0/24 is directly connected, FastEthernet0/0
L    10.2.2.2/32 is directly connected, FastEthernet0/0
O E2 192.168.14.0/24 [110/20] via 10.2.2.3, 00:19:02, FastEthernet0/0
O E2 192.168.34.0/24 [110/20] via 10.2.2.3, 00:19:02, FastEthernet0/0
O E2 192.168.44.0/24 [110/20] via 10.2.2.3, 00:19:02, FastEthernet0/0
Area1R2#
```

5.3 Floating Static Route Redundancy

```
CoreR1
!
!
!
!
!
interface FastEthernet0/0
 ip address 192.168.14.1 255.255.255.0
 duplex half
!
interface GigabitEthernet1/0
 ip address 10.1.1.1 255.255.255.0
 negotiation auto
!
interface GigabitEthernet2/0
 ip address 10.1.10.1 255.255.255.0
 negotiation auto
!
router ospf 1
 network 10.1.1.0 0.0.0.255 area 0
 network 10.1.10.0 0.0.0.255 area 0
!
ip forward-protocol nd
no ip http server
no ip http secure-server
!
!
ip route 10.1.10.0 255.255.255.0 GigabitEthernet2/0
ip route 10.1.10.0 255.255.255.0 FastEthernet0/0 50
!
no cdp log mismatch duplex
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
```

5.4 Data Plane Connectivity and Verification

```
BranchR4
*Jul 14 13:10:50.695: %SYS-5-CONFIG_I: Configured from memory by console
*Jul 14 13:10:51.503: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.3(3)XB12
, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Tue 19-Nov-13 04:39 by prod_rel_team
*Jul 14 13:10:51.615: %SNMP-5-COLDSTART: SNMP agent on host BranchR4 is undergoi
ng a cold start
*Jul 14 13:10:51.943: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Jul 14 13:10:51.947: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
BranchR4#
BranchR4#Show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 3 subnets
R    10.1.1.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.1.10.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.2.2.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/0
L    192.168.14.4/32 is directly connected, FastEthernet0/0
192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet1/0
L    192.168.34.4/32 is directly connected, GigabitEthernet1/0
192.168.44.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.44.0/24 is directly connected, GigabitEthernet2/0
L    192.168.44.1/32 is directly connected, GigabitEthernet2/0
BranchR4#
```

```
BranchR4
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 3 subnets
R    10.1.1.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.1.10.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.2.2.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/0
L    192.168.14.4/32 is directly connected, FastEthernet0/0
192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet1/0
L    192.168.34.4/32 is directly connected, GigabitEthernet1/0
192.168.44.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.44.0/24 is directly connected, GigabitEthernet2/0
L    192.168.44.1/32 is directly connected, GigabitEthernet2/0
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#ping 10.1.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 136/143/152 ms
BranchR4#
```

```
BranchR4
10.0.0.0/24 is subnetted, 3 subnets
R    10.1.1.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.1.10.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
R    10.2.2.0 [120/5] via 192.168.34.3, 00:00:10, GigabitEthernet1/0
192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/0
L    192.168.14.4/32 is directly connected, FastEthernet0/0
192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet1/0
L    192.168.34.4/32 is directly connected, GigabitEthernet1/0
192.168.44.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.44.0/24 is directly connected, GigabitEthernet2/0
L    192.168.44.1/32 is directly connected, GigabitEthernet2/0
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#
BranchR4#ping 10.1.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 136/143/152 ms
BranchR4#
BranchR4#
BranchR4#traceroute 10.1.10.1
Type escape sequence to abort.
Tracing the route to 10.1.10.1
VRF info: (vrf in name/id, vrf out name/id)
 1 192.168.34.3 92 msec 92 msec 76 msec
 2 10.2.2.2 64 msec 40 msec 72 msec
 3 10.1.1.1 88 msec 92 msec 92 msec
BranchR4#
```

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6. Conclusion

The successful deployment and testing of the Enterprise Multi-Protocol Routing Challenge demonstrate a fully converged, highly resilient network architecture. By effectively separating the link-state boundaries across OSPF Area 0 and Area 1, control plane overhead and database flooding are minimized, ensuring high performance throughout the corporate backbone. Furthermore, the integration of the legacy branch through mutual OSPF and RIPv2 route redistribution on the ASBR gateway establishes flawless end-to-end reachability without compromising security or architectural integrity. Backed by a floating static route configuration for dynamic path redundancy, the implementation stands verified, fully operational, and compliant with enterprise network standards.